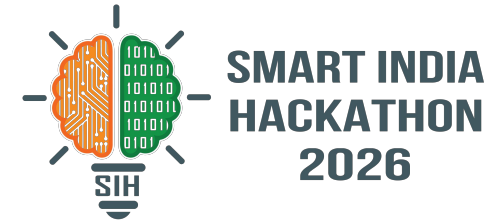
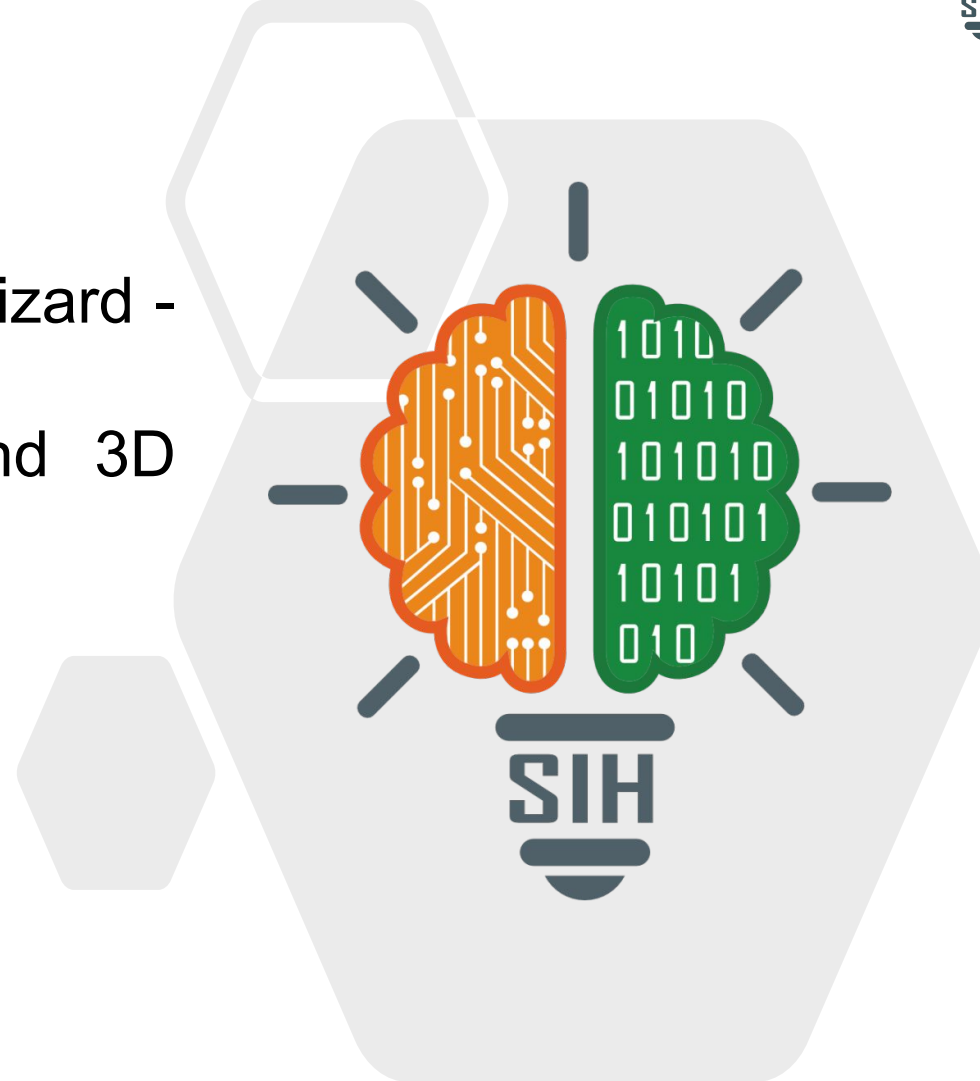


SMART INDIA HACKATHON 2026



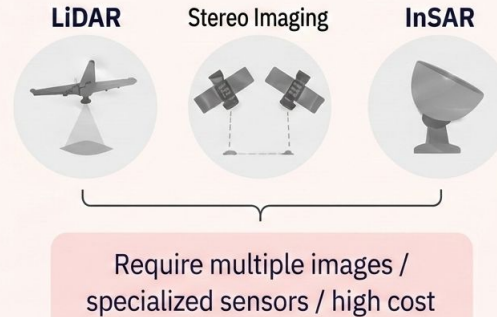
- **Problem Statement ID** – 26175
- **Problem Statement Title**- DepthWizard -
Single-View Height Estimation and 3D
Flythrough
- **Theme**- Disaster Management
- **PS Category**- Software
- **Team ID**- 188947
- **Team Name**- Team Nexus_theem



IDEA TITLE

Problem

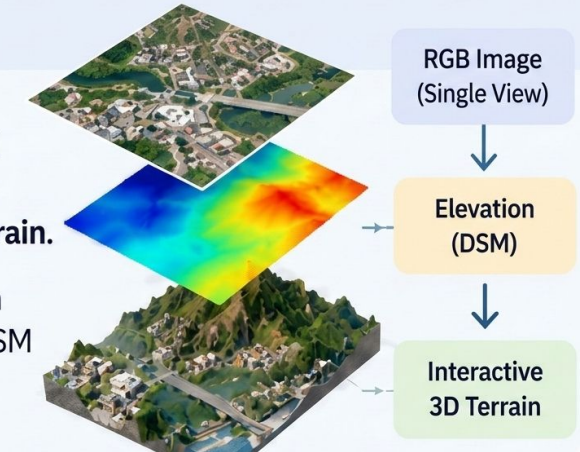
Accurate elevation data usually requires **LiDAR, stereo imaging or InSAR**, while single-view RGB images provide only **scale-ambiguous relative depth**, making elevation reconstruction and interactive 3D analysis challenging for remote-sensing scenes.



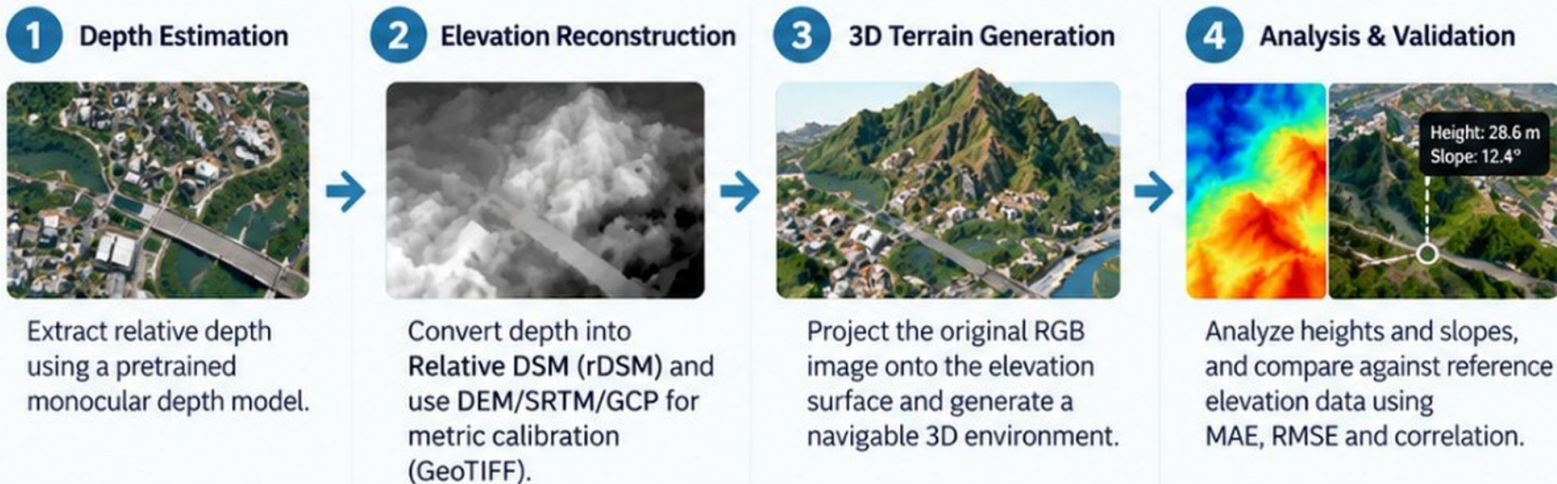
Our Idea

DepthWizard transforms a **single optical remote-sensing image** into an **elevation representation** and an **interactive 3D terrain**.

It supports relative DSM generation from normal images and a calibrated metric-DSM workflow for georeferenced imagery.



Proposed Solution



Innovation & Uniqueness

-  **Calibration-Aware Pipeline**
Explicit relative-to-metric elevation workflow for georeferenced imagery (DEM/SRTM/GCP).
-  **Calibration Experimentation**
Compare different scale-calibration strategies across scene conditions.
-  **Terrain-Wise Validation**
Evaluate reconstruction across urban, sparse, hilly and forested scenes.
-  **Error-Aware 3D Analysis**
Connect quantitative spatial errors with interactive terrain and height/slope analysis.

Technology Stack



Frontend

- **React + TypeScript + Tailwind CSS**
Web dashboard and analysis interface
- **Three.js / React Three Fiber**
Interactive 3D terrain and flythrough



Backend

- **Node.js + Express.js**
APIs, file handling and processing orchestration
- **MongoDB**
Projects, experiments and validation results



AI & Image Processing

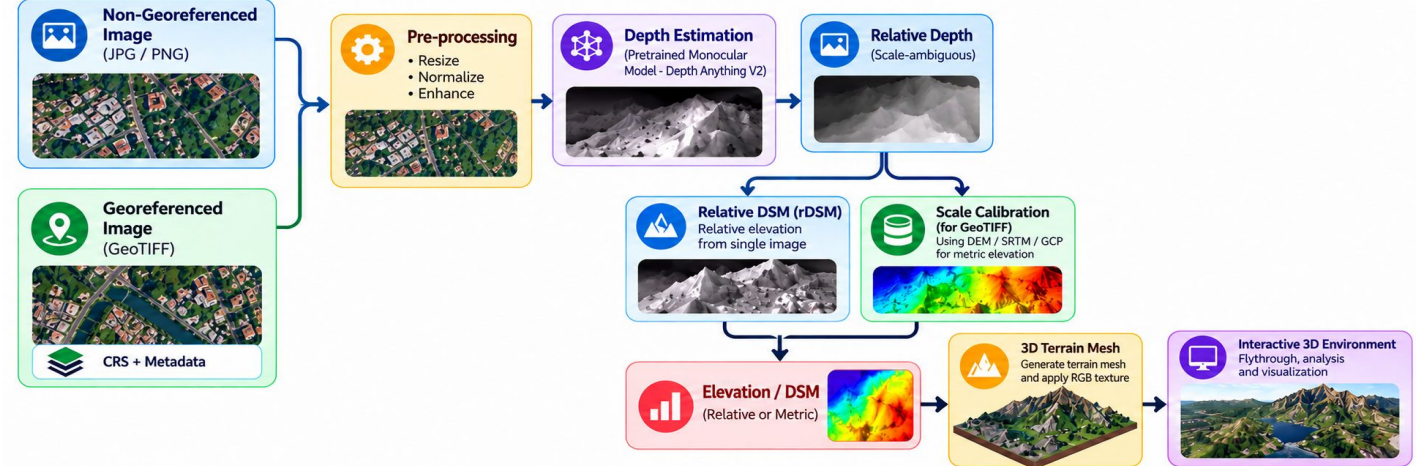
- **Python + FastAPI**
AI and scientific processing service
- **PyTorch (Depth Anything V2)**
Monocular depth estimation
- **OpenCV + NumPy**
Image and depth processing



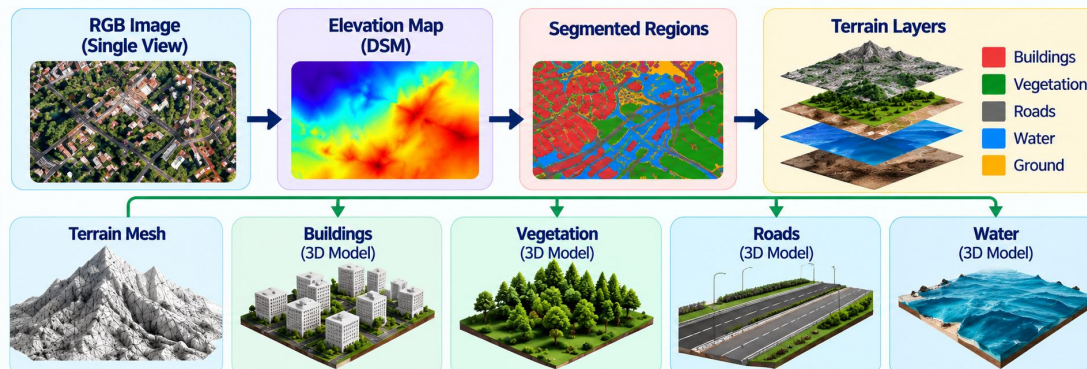
Geospatial Processing

- **Rasterio / GDAL**
GeoTIFF, CRS and elevation data processing
- **SRTM / DEM / GCP**
Scale calibration for metric elevation

System Workflow



3D Environment Generation



Github Link:

<https://github.com/NirajPatil333/TerraVision.git>

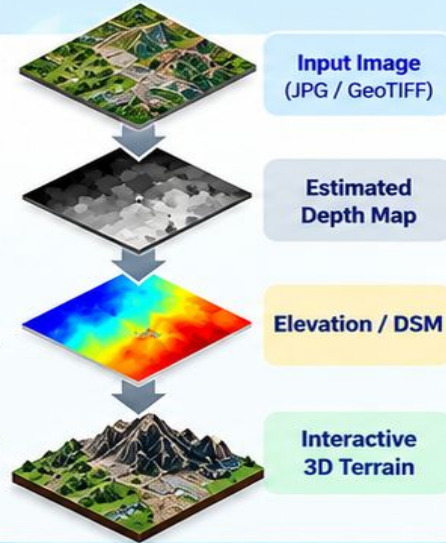
Report Link: [DepthWizard](#)

Demo Video:

<https://youtu.be/nAdUAiqz5YU?si=5saNXFizYq-V6IE->

Feasibility

- **Software-only solution:** Works with standard satellite/drone images, no specialized hardware required.
- **Pretrained models:** Uses existing monocular depth models (e.g., Depth Anything V2), no model training needed for MVP.
- **Publicly available datasets:** Open satellite imagery and reference elevation data (e.g., SRTM, DEM, LiDAR) for development and validation.
- **Modular architecture:** MERN stack for application layer and Python for AI/processing enables independent and parallel development.
- **Dual input support:** Works with both non-georeferenced images (JPEG/PNG) and georeferenced images (GeoTIFF) for relative and metric workflows.



Commercial Feasibility / Viability

- **Market Demand:** Growing need for 3D geospatial visualization in urban planning, disaster management, environment monitoring and infrastructure planning.
- **Cost-Effective:** Uses open-source models and publicly available datasets, reducing development cost.
- **Accessible:** Web-based platform can be used by researchers, government agencies and other stakeholders.
- **Scalable:** Can support additional data sources, higher resolution imagery and advanced calibration methods.
- **Future Scope:** Can be extended to support time-series analysis, change detection and custom domain applications.



Urban Planning
3D visualization for smart cities



Environmental Monitoring
Forest, land and terrain analysis



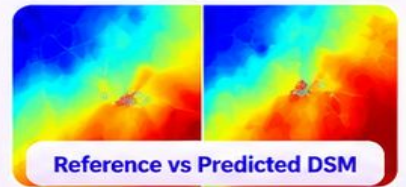
Disaster Management
Flood, landslide and damage assessment

Challenges

- **Relative depth is scale-ambiguous:** Single-view images do not provide absolute metric elevation.
- **Remote-sensing domain gap:** Performance may vary across different terrain types (urban, sparse, hilly, forested).
- **Reference data dependency:** Metric validation requires DEM/LiDAR/GCP-based data.
- **High-resolution processing:** Large satellite images and 3D meshes require significant computational resources.
- **Variations in image quality:** Cloud cover, shadows and low-resolution images can affect depth estimation.

Strategy / Mitigation Plan

- **Scale Calibration:** Use DEM (SRTM) / LiDAR / GCP to convert relative depth to metric elevation.
- **Terrain-wise Evaluation:** Test across different environments – urban, sparse, hilly and forested.
- **Model Optimization:** Use high-resolution tiling and mesh simplification to reduce computation cost.
- **Multiple Calibration Methods:** Compare and select suitable methods for different terrain conditions.
- **Error Analysis:** Quantitative validation using MAE, RMSE and correlation with reference DSM/LiDAR.





Target Audience & Impact



Disaster Management Authorities

- Quick terrain assessment from available satellite/drone images.
- Helps in flood, landslide and terrain risk analysis.
- Supports planning and rescue operations.



Urban & Regional Planners

- Interactive 3D terrain for urban planning and infrastructure analysis.
- Helps evaluate terrain conditions, slopes and structural heights.
- Supports smart city and land-use planning.



Survey & Remote-Sensing Teams

- Generates elevation representations from single optical images.
- Useful when dedicated elevation acquisition is unavailable.
- Supports field planning and preliminary analysis.



Researchers & Analysts

- Compare reconstruction and calibration methods.
- Evaluate performance across different terrain conditions (urban, sparse, hilly, forested).
- Useful for academic research and geospatial analysis.



Broader Impact



Better Decision-Making

Provides terrain insights for disaster response, planning and resource allocation.



Improved Accessibility

Makes elevation and 3D terrain information available to wider users through a web platform.



Supports Sustainability

Assists in environmental monitoring, land management and conservation.



Enables Research & Innovation

Provides a foundation for advanced geospatial analysis and new applications.



Key Benefits



Operational Benefits

- Single-image workflow for quick terrain visualization.
- Interactive 3D analysis with height, slope and terrain inspection.
- Dual input support — JPG/PNG for relative DSM and GeoTIFF for metric workflow.



Economic Benefits

- Reduces dependence on specialized data acquisition for initial terrain assessment.
- Uses pretrained models and publicly available datasets.
- Software-based solution avoids dedicated hardware requirements.



Environmental Benefits

- Helps in terrain and land surface analysis using existing optical imagery.
- Supports preliminary assessment of flood, landslide and terrain risk areas.
- Useful for forest, land and watershed analysis.



Real-World Applications



Flood & Landslide Assessment



Urban & Land Management



Terrain & Land Analysis



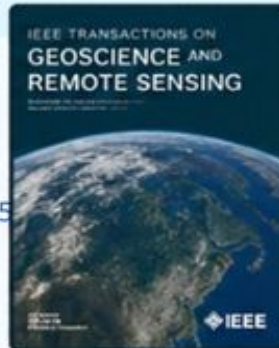
Environmental Monitoring

1 Base Research Paper

Depth2Elevation: Scale Modulation With Depth Anything Model for Single-View Remote Sensing Image Height Estimation

Zhangcheng Hong, Tong W, Zhiyan Xu, Wufan Zhao
IEEE Transactions on Geoscience and Remote Sensing (TGRS), 2025

DOI: [10.1109/TGRS.2025.3564820](https://doi.org/10.1109/TGRS.2025.3564820)



Relevance: Proposes single-view remote-sensing height estimation using Depth Anything with scale modulation. Provides insights into scaling relative depth to metric elevation and evaluation on real-world datasets.

2 Technology Foundation

Depth Anything V2: A Robust Foundation Model for Monocular Depth Estimation

Lihe Yang et al.

NeurIPS 2024

<https://arxiv.org/abs/2401.10891>

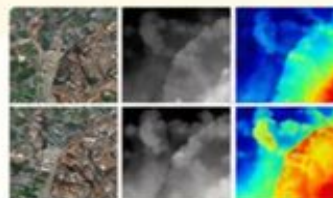


Relevance: Pretrained monocular depth estimation model used in our system to generate relative depth from a single RGB image, with strong performance across diverse scenes.

3 Dataset / Benchmark Reference

GAMUS: Benchmark for Remote Sensing Height Estimation

Guo et al.
2024 / 2025 (Referenced in Depth2Elevation)
<https://github.com/zhongchenghong/GAMUS>



Relevance: Public benchmark dataset used for evaluating single-view remote-sensing height estimation methods. Useful for terrain-wise validation and comparison.

Research Gap and Our Direction

Existing Research

Single-view depth estimation and height estimation from remote-sensing images.

Research Gap

- Limited work on complete end-to-end platform.
- Scale calibration and metric DSM generation not fully explored.
- Lack of integrated 3D interactive analysis tools and validation framework.

Our Research Direction (DepthWard)

- Build a complete pipeline from RGB image to elevation and 3D terrain.
- Support both georeferenced and non-georeferenced workflows.
- Integrate scale calibration, DSM reconstruction and validation.
- Provide an interactive 3D environment with height and slope analysis.

Input Image
(Single Optical Image)



Estimated Depth Map
(Using Depth Anything V2)



3D Terrain Visualization
(Our Platform)



Additional References

- Monocular Depth Estimation for Remote Sensing: A Survey
Zhang et al. (2023) <https://doi.org/10.1109/TGRS.2023.xxxxx>
- Digital Surface Models and Elevation Extraction from Optical Imagery
Li et al. (2022) <https://doi.org/10.3330/rsi14717418>
- Open Elevate Data Sources (SRTM)
USGS Earth Explorer | <https://earthexplorer.usgs.gov/>